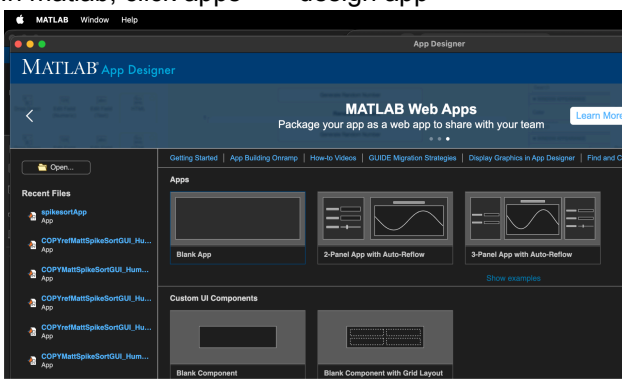
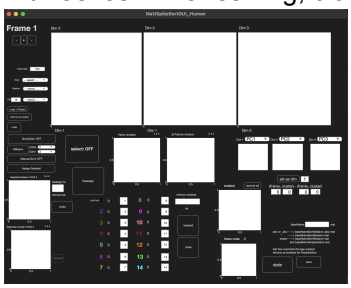
APP INPUT: RawSortDat_subj_sess_ch_Thresh.mat, which is the output from the extraction code
APP OUTPUTs (2): RawSortDat_subj_sess_ch_Thresh.mat (by default, the same exact file as the raw extraction output, but with more variables) and AnalysisVars_subj_sess_ch_Thresh.mat (used for decoding)

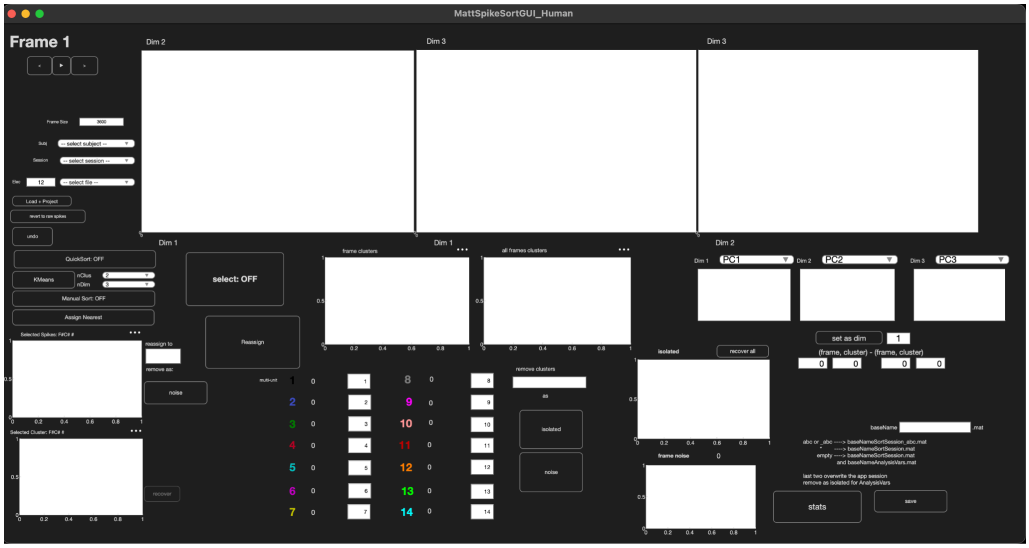
In matlab, click apps —> design app



Click open and locate the app file:
'.../Nelson Lab Projects
/Spike Sorting pipeline project
/SpikeSort Matlab
/spikesortApp.mlapp'
***MAKE SURE IT IS .mlapp

Go to code view
Change DataRoot char = ‘pathto/Nelson Lab Data/Intracranial data/Raw Neural Data per Subj/’
Run
If it freezes while resizing, close, reopen, and wait a moment before resizing.





Plots (explained further below)

Top:

three main PC plots

Bottom left:

selected spikes plot (top), selected cluster plot (bottom)

Two medium plots below the big three:

mean waveforms for all clusters in current frame (left)

mean waveforms for all clusters across all frames (right)

Between those two medium plots and the little three:

mean waveforms for every isolated cluster across all frames (ie, one mean waveform for all spikes assigned to cluster 3 over all frames) (top)

mean waveform for every 'noise number,' noise numbers just count the remove-as-noise-events and can be thought of as noise clusters

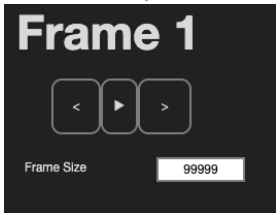
Frames divide the spike file into time-segments, which makes sorting easier

Frame size is in seconds and can be changed within a session without reloading the data

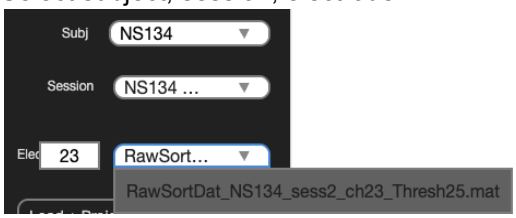
This is useful for comparing the stats (explained below) for parts of/the whole file

By default, everything is in one frame (ie 9999) which is ideal for sorting

Note: the exported AnalysisVars (explained below) does not include frame information



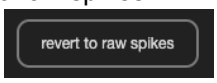
Select subject, session, electrode



When you type electrode, all RawSortDat files from that subject/session/electrode will load in the electrode dropdown

Once file selected, 'Load + Project' projects onto the PC plots (explained below)

Revert to raw spikes

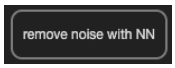


Retrieves all spikes and puts them in cluster 1

Undo: works for anything, can also do cmd/cntrl Z



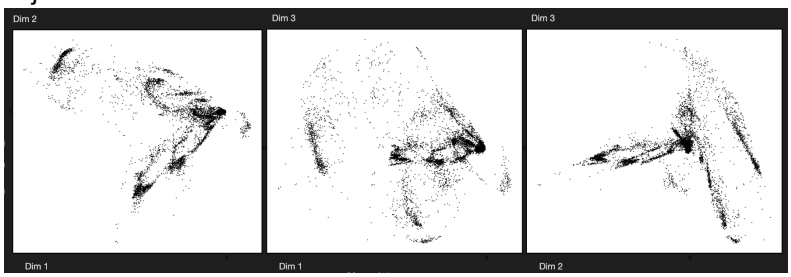
Remove noise with NN (neural network)



See code for training the neural network [it is trained on all not-raw RawSortDats (i.e. RawSortDats that have been opened in the app, and where spikes have been assigned clusters) from a given subject]

The button uses the neural network to detect what it considers noise and move those spikes to the next available cluster

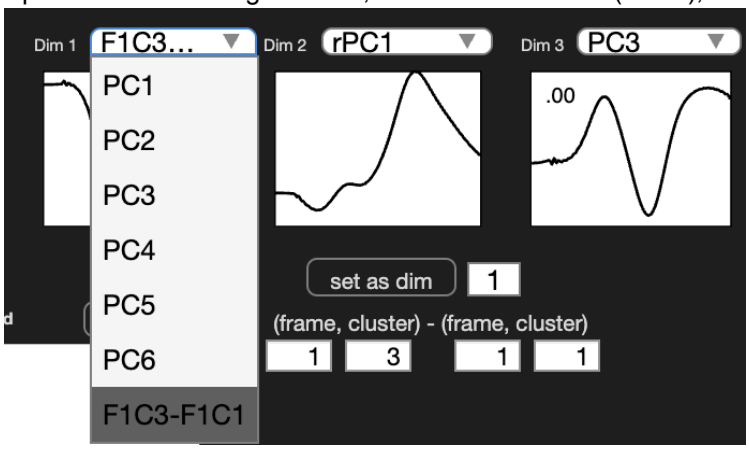
Load + project



Projects onto PC plots and calculates PCs

The computed PCs become the default plot dimensions

3 plot dimensions: regular PCs, custom dimensions (F#C#), and residual PCs (rPCs)



Plotting dimensions

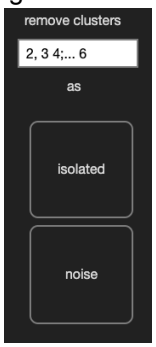
Note: ‘Dim 1-3’ are not always PCs, they are just what is plotted

PCs, custom dimensions (F#C#), rPCs (residual PCs, only possible with a custom dimension)

PCs are recomputed any time spikes are removed (as noise or as isolated, see below)

Note: PCA and everything in these dropdowns (except for custom dimensions which can be used in any frame, see below) are computed per-frame. So each frame has different PCs

Removing clusters as noise or as isolated removes them from the frame’s ‘PC space’

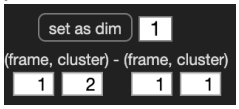


The PCs are recomputed

The big plots change since the plot dimensions were updated

Important: for removed-as-isolated clusters, you cannot add spikes to them, or send them or any spikes from them into other clusters. To change anything about a removed-as-isolated cluster, recover the entire cluster first.

Custom plot dimensions



Sets the mean waveform of a cluster (or the difference of 2) as the chosen plot dimension

Leave the ‘-(frame, cluster)’ at 0 0 if you don’t want a difference

rPCs / residual PCs

Anytime a custom dim is selected (except for when custom dims are selected for all 3 dims), 3 residual PCs are computed (explained below)

Those residuals PCs are automatically selected to fill the rest of the Dim options from left to right

Compute rPCs

Project non-removed spike waveforms onto the plotDims to the left of the custom dim

Subtract that projection from those spike waveforms —> now we have residuals

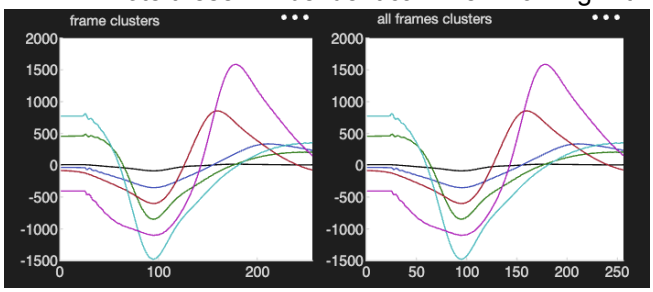
Do PCA on those residuals —> rPCs

Two plots below the big three:

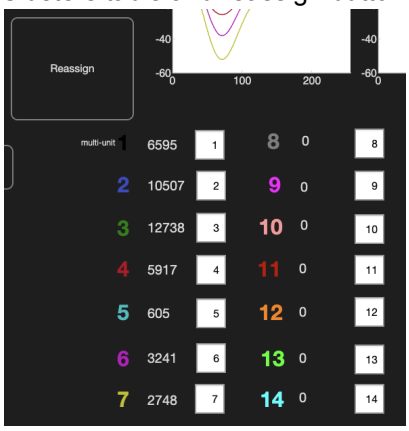
Left: mean waveform for each cluster in the current frame

Right: mean waveform for each cluster in every frame

*Note these will be identical when working with one frame



Clusters table and reassign button

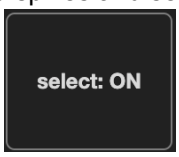


White text = number of spikes in cluster (regarding the current frame only)

Type the cluster# you want to reassign the spikes to and click Reassign

This does NOT recompute PCs

Selected spikes and selected cluster



Always double click to disable

Uses:

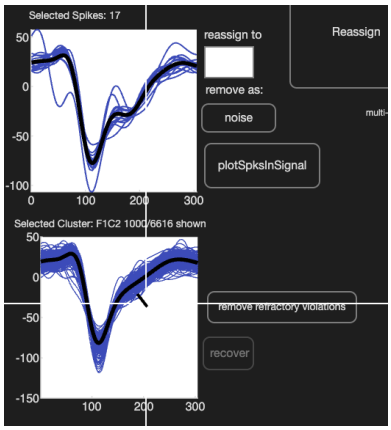
Hover over spikes on the PC plot and display them on the selected spikes plot

Select a mean waveform from either mean waveform plot to plot a random 1000 of its spikes in the selected cluster plot

Hover over spikes on the selected cluster plot and display them on the selected spikes plot

Select a noise number and plot the noise ‘cluster’

Draw a line on the selected cluster plot to select a subset of spikes from the plotted cluster



You can then send the selected spikes to any cluster or remove them as noise.

Note: since this is NOT an isolated cluster, the ‘recover’ button is available.

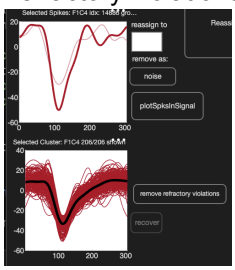
Recover button

For recovering an entire isolated cluster.

For noise ‘clusters’, recovering sends the to the mutli-unit cloud, 1 (black)

The noise button does nothing if they are already noise waveforms

Remove refractory violations (see stats section for explanation of refractory violations)

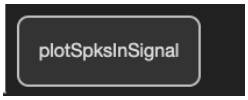


When pushed, sends all of the spikes within 3ms of any other spike to the next available cluster

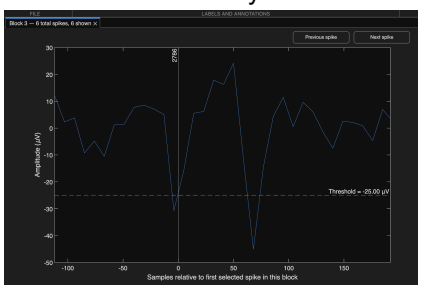
Note: It does not just send the first or just the second spike within the 3ms interval. Rather it sends all the spikes. You can still hover over the plot and select normally. The transparent waveforms on the selected plot are the spikes that occurred within 3ms of the selected (non-transparent) spike.

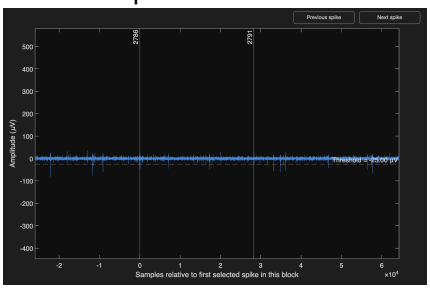
Result: the ‘refractory’ cluster will have a refractory index of 100% (containing only the violating spikes) and the cluster that was selected when the button was pushed will have a refractory index of 0%.

plotSpksInSignal

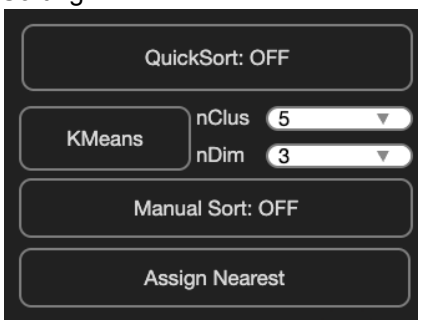


For your selected spikes, plot them in the ‘detection signal’ (the signal that detection was run on) This helps for distinguishing spikes from noise. Often for noise it is periodic.
See the function plotSpksInSignal to edit the max amount of spikes per block and other settings
WARNING this is very slow and needs to be improved





Sorting



For all sorting options

Clusters that are already removed as isolated are ‘reserved.’ Sorting methods will use the next available ones.

Quicksort and manual sort:

- Quicksort
  - Place dots on the 3 main plots, each spike is assigned to the nearest dot
  - Nearest = squared Euclidean distance in the 2-d plotted space ONLY
- Manual sort
  - Draw boundaries around spikes
- Both: right-click places the last object AND executes the sort

KMeans (also see Stats + Methods google doc for more in-depth explanation)

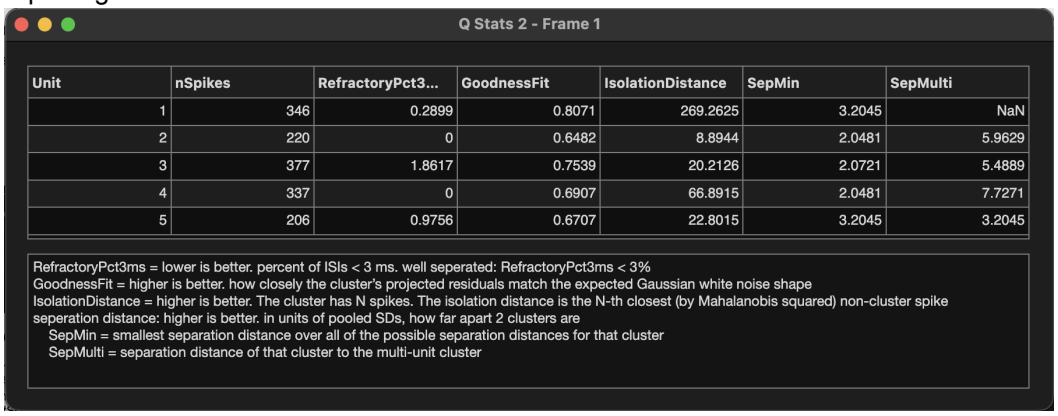
- Use the projections onto the first nDim plotting dims
- Creates nClus random points
- Assigns spikes in the current frame to the nearest point (squared Euclidean in nDim space)
- Compute new points as: the mean of the points assigned to each original random point
- Assign using distance again, and repeat
- Parameters (actual code names)
  - Replicates = number of times to do the entire KMeans looping process, where the best result is chosen out of them
  - MaxIter = max cycles per ‘replicate’

Assign nearest

- Assign each non-removed spike in the frame to the nearest already-existing cluster mean
  - Squared Euclidean in nSamples space (Wfs matrix, which is nWaveforms x nSamples)
  - Will not create new clusters

Stats

Computed per frame. Change the framesize to ie 9999 to see whole-session stats before exporting.



The count on top (Q Stats #) increments for each stats window in the sort session, to compare easily before/after doing an action. These windows stay open until closed.
Metric meanings. See Stats + Methods google doc for more in depth explanations.
Some metrics cannot be computed and show NaN for small clusters
SepMulti is always NaN for unit 1 since it is the multi-unit cluster
Most important metric is refractory period which should be less than 3 ideally

Saving



Import RawSortDat_subj_sess_ch_Thresh.mat
Text bar will effect any save:

If text bar has *, RawSortDat is overwritten, with no analysisvars saved

If text bar is empty, the RawSortDat is overwritten AND an analysisvars is saved

If text bar contains writing, the RawSortDat will be duplicated and have your writing appended ie RawSortDat_abc

You can always load a sort session, revert to raw spikes, save the session and change the name to CSC##spikes.mat to effectively get the spikefile back

The data structure will be different, but it makes no difference for sorting